BIO144 Course Information (version 2026)

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2026-01-23

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Preface

This website contains information about the BIO144 Data Analysis in Biology course at the University of Zurich.

Beware that Owen sometimes makes updates to the information during the semester, so if you have downloaded a copy or taken screenshots, your copy may not exactly match the most current version. However, any changes will be correcting typos or improving explanations. If content does change in an important way Owen will announce this in the lectures, or on OLAT, or by email.

How this website was made

The book was written using a type of RMarkdown. It allows a script with a mix of normal text and R code to produce chapters and a book that has a mixture of text, R code, and R output. Rmarkdown is very useful for making reports, books, presentations, and even websites.

This book is a Quarto book. To learn more about Quarto books visit https://quarto.org/docs/books.

General information

Aims and learning outcomes

This course will help you develop a solid foundation in answering biological questions with quantitative data. Moreover, the approaches you will learn are generally applicable to using data to solve problems, an increasing important skill in a world more and more dominated by data.

The Learning Objectives of the course tell you what you need to learn and know. Hence, it is very important for you to be very clear about the learning objectives of the course. They will help you know what you're expected to learn and what you will be examined on. If you can achieve all of the learning objectives, you will be very well prepared for a great future in data analysis (and for the exam).

Course schedule

The course runs during the 14-week spring semester. The course has 12 weeks of new content, and 2 weeks of self-study (no new content). Each of the 12 weeks of new content has a set of learning objectives. Each week has four parts: a lecture, homework, a practical session, and a practice quiz.

Lectures are on Monday mornings, 8:00 - 9:45. Practical sessions are on Thursday or Friday afternoons, 13:00 - 14:45.

You'll be assigned to a practical session on either Thursday or Friday. If at any point you wish to change afternoons permanently, contact `studienkoordination@biol.uzh.ch`. If you want to swap days for just one or two weeks, please do so without asking.

Please see the "Year specific information" section below for a detailed course schedule.

What should I already know?

Please see the section in OLAT "Previous Knowledge".

Expected workload

The course is worth 4 credit points, so with 1 CP being approximately 30 hours work, requires around 120 hours study. Approximately one third of this is "in class" and two thirds is self-study, including in-course assessments and preparation for the examination.

Bring your own laptop

Please bring your laptop to class. There are some power outlets in the lecture theatre, but best will be to make sure that your laptop has sufficient battery charge to last the duration of the class (up to 2 hours), and consider using an external additional battery (powerbar?) if required. Please make sure you know how to use your laptop (some people are sometimes not so sure, perhaps because its new, or they borrow someone else's.) Your laptop must be able to connect to the wireless internet at UZH.

We will be using three computer programmes during class (as well as the usual suspects, such as an internet browser etc.) please make sure you have them:

1. A spreadsheet programme, such as Microsoft Excel, Mac Numbers, or OpenOffice.
2. The R software environment for statistical computing and graphics. It's worth knowing why we are using R. You can find out by reading Section 4 of the Preface of the Beckerman, Childs, and Petchey Book.
3. Rstudio. We use RStudio to interact with R. Rstudio has lots of features that make working with R much much easier and enjoyable.

If you need any help setting up your laptop with R and RStudio, please look at relevant section of the "Getting R & RStudio" section of the "Previous Knowledge" section of the course. If you still have problems, ask during the first practical.

What to do each week

- Attend the lecture.
- After the lecture and before the practical do the homework.
- Come to and do the practical.
- Do the practice quiz.
- Have fun ;)

Course texts

The required content of the course is all available in the course book: BIO144 Course Book. The lectures will help you understand the content of the book. The homework and practicals will also help you learn the material. Practice quizzes will help you check your learning, and prepare for the final exam. All of this will also help you develop your skills in R, which will also be required in the final exam.

There is other optional reading that we may suggest to you during the course. This is listed below. Textbooks for optional reading are in the main library, and should be accessible electronically through the main library.

Optional reading comes from the following books. Also each chapter of the course book suggest reading to further your understanding.

- *Linear Regression, Seminar für Statistik, ETH Zürich*(2008 / 2013) Werner Stahel. And here is an English version of that text.
- *Getting Started with R, An introduction for biologists* (2017, Second Edition) Beckerman, Childs & Petchey. (**DO NOT USE THE FIRST EDITION!**). The link is to the book on the Oxford University Press website; full access to the book will require you have a UZH ip address (i.e. are on the UZH network or VPN into it).
- Insights from Data with R, by Petchey, et al. (Same access information as the previous book in this list).
- *The New Statistics with R, An Introduction for Biologists* (2015, First Edition) Hector. Ebook via UZH library.

Some other nice (though optional) texts that you might find interesting.

- *Regression; Models, Methods and Applications* (2013 Edition) Fahrmeir, Kneib, Lang & Marx.

- *The Analysis of Biological Data* (2015, Second Edition) Whitlock & Schluter.

- The Essential Guide to Effect Sizes. Statistical Power, Meta-Analysis, and the Interpretation of Research Results (2010, First Edition) Ellis. Ebook via UZH library.

- *Statistics: An Introduction using R* (2011, Second Edition) Crawley.

Getting datasets

You can get the datasets used in the course here. To download one dataset from that web page, click on the dataset and you then see a webpage with that dataset. Then right click on the “Raw” button, and select “Save As”, select location to save to, and press Save. Alternatively you can get a zip file of all the datasets here, which you will then need to unpack before use.

You can get datasets associated with the book Getting Started with R, by Beckerman, Childs, and Petchey here.

IMPORTANT: if you open the downloaded file in Excel, perhaps to have a look at it, then ***DO NOT, UNDER ANY CIRCUMSTANCES, ASK OR ALLOW EXCEL TO SAVE IT***. (Excel does not like R and will do everything in its power to mess things up for R. Don't let it do that.)

Here is the data you collected during the first lecture:

human_benchmark_data_fs25.csv

Really important!!!

Please please please read, understand, and try your very best to remember what is below.

We will very often be importing data into R using the read.csv or read_csv function. In the time since the material of the course was written (including texts such as Getting Started with R) the makers of R changed something important about how importing data works. The best solution is for us to make sure we change it back to the old way, so all the course material makes sense!

Here are two things you must always do:

When using read.csv always add the argument 'as.is = FALSE.
For example: dd <- read.csv("my data file.csv", as.is = FALSE)

When using read_csv, always pipe the result into mutate_if(is.character, factor).
For example:
dd <- read_csv("my data file.csv") %>%
mutate_if(is.character, factor)

Attendance

Attending class is a really good idea!: "attendance [is] a better predictor of college grades than any other known predictor of academic performance", Credé, Roch, & Kieszczynka, 2010. The full article.

We give you the freedom to use the learning materials and opportunities in the way that suits you best. We strongly believe that attending the practical classes and lectures will be an efficient, fun, and valuable way for you to learn.

We will not be tracking attendance. If you do not come to class (practical or lecture) the only penalty you will potentially experience is loss of some learning opportunity.

What if I was ill or otherwise unable to attend? Please use your initiative to catchup. Lectures will be available as podcasts. Ask for help from friends for help with practicals. And whatever else you can think of.

Self study weeks

Consolidate, review, reinforce, expand!

During approximately two weeks there is no lecture and no practical. During these week you should take the opportunity to consolidate, review, reinforce, expand!

Here is some optional reading you could look at during those weeks:

- Retire Statistical Significance
- Interleaf from Whitlock and Schluter: Statistical significanc vs. biological importance
- Interleaf from Whitlock and Schluter: Correlation vs. causation
- P-Werte in der NZZ (3. April 2016) (You will have to zoom in and probably read it on the screen!)
- And finally, a really thoughtful article in one of the world's leading scientific journals: S. Goodman (2016): Aligning statistical and scientific reasoning

Assessment

There will be two assessment-related types of activity:

1. Weekly practice quizzes

These are only for you. Your scores do not contribute to anything. The are intended:

- To make sure you know if you're learning and understanding the material.
- To motivate you to keep up with the content of the class, and thereby reduce the chance of you falling behind and then having trouble in the final exam.
- To give you practice with the types of questions that will be in the final examination.

There is no deadline for the practice quizzes. You can do them whenever you like. You can do them multiple times if you like.

2. The final exam (and the resit exam)

A final assessment during the module examination period. Your performance in this is the only determinant of your final grade.

Examination date and time: Please see the Studium Biology, Modulprufungen im Grundstudium web page and the links therein for the examination dates and other important examination information.

This information is draft for 2026 and may still be changed.

Further information:

- The exam will contain about 40 questions. Each will be one of three types: single choice, multiple choice, or numeric. Single choice is a question with several possible answer, of which only one is correct – you indicate which. Multiple choice is a question with several possible answers; none, 1, 2, 3, or 4 may be correct – you indicate which. Numeric requires you to enter a number to answer the question.
- A full mock exam will be available on OLAT.
- During the exam you are allowed **three sheets of A4 paper** that you may put information on both sides of. You are also permitted to bring a dictionary (it must have no writing in it). No other resources are permitted.
- You are allowed to bring into the exam one translation dictionary (Übersetzungswörterbuch) (e.g., an Englisch–Deutsch-Wörterbuch / English–German dictionary). Examination staff should be allowed to inspect this resource.
- You will need to present your UZH card during the exam, to verify your identity.
- You will need to use your laptop continuously during the two hours of the exam. Some power outlets are available, but best is to make sure you have enough battery power to last the exam. Please ensure your laptop can connect to the internet via wifi.
- You will take the exam in a special instance of OLAT from within the *Safe Exam Browser* that must be installed on your computer.
- From within the Safe Exam Browser you will be able to use an RStudio Server.
- There will be a mandatory practice exam experience including use of the Safe Exam Browser. Details will be sent to you.
- The exam is in English.
- You must agree with an Honour Code on OLAT immediately before you take the exam.
- You will sit the exam in a lecture theatre on the Irchel Campus.
- The final exam requires you to analyse some datasets that are provided to you. Some students have reported that they didn't have enough time during the exam, so please ensure that you can do things like importing data quickly and easily.
- Your working (e.g. any R code your write) is not assessed directly. It is assessed indirectly by whether you answer a question correctly.
- During the exam, if you have any trouble importing the data, please immediately ask for help.

- When answering numeric questions, the default is to give your answer with precision of 2 decimal places. Some questions specify to give the answer to a different number of decimal places. Please be careful to take care of this.

Important information about the "practice exam experience"

The practice examination experience is only to give you experience of the examination interface, the types of questions that can be asked, and to test the examination interface. Do not use the number and type of different types of question as an indicator of anything about the final exam.

Here are the questions and solutions that are used in the practice exam experience.

You will be informed by email when the practice exam experience is, and all further details.

Getting help

We aim that you can ask for help about anything and get a useful answer as soon as possible. The following guidelines will help you get fast, useful help. The primary ways to ask for help are:

1. During the practicals on Thursday and Friday. You will get close to instant responses.
2. Anytime on the Discussion Forum of this OLAT course. We aim that you get a response in less than two days. Anonymous posts are possible.
3. After lectures.

Of course, please feel free to email Owen directly if you wish. Please don't hesitate to ask for help!

Posting code problems on the Discussion Forum

The Wiki:FAQ contains a page ***How to ask a question so we can easily help***. Please take a look.

Specialist statistical assistance

I (Owen) am not a statistician. Hence, I might be unable to answer some of your questions about more statistical issues, particularly as we experience more and more complex statistics. To deal with this, Prof. Furrer has agreed to help with about statistics that I cannot answer (or cannot answer fully). Please post them as a new thread in the discussion forum. Include as much information as possible, including R code and output as appropriate. Please do not contact Prof. Furrer directly. I will then contact Prof. Furrer and feedback the answer to you.

Giving feedback

Your views on the course are really important to us. What works well, what doesn't work so well, what content is particularly important, what content is not so important, what content is missing?

There are several ways of providing feedback.

1. During the course, you can tell us in person during class, and can make posts on the Forum. Such feedback during the course has the potential to improve the course for you. I.e., we might be able to change something quickly, so your and your fellow students' experience of the class is improved.
2. Fill in the "official" course evaluation form. Details to follow.
3. If have anonymous feedback to give us before or after the official course evaluation then please write on some paper and post it through the University internal mail to us.

Course language

The course will be mostly in English. Many of the TAs speak and understand English and German. Prof. Petchey best understands and speaks English.

Please let us know how we can assist with any challenges this presents.

Here is a web page that gives most common (and many not so common) statistical terms in a multitude of languages!

Reference letters

The course instructors and the TAs try to get to know each of you during the course, particularly in the practical classes. That said, it is unlikely we gain enough experience of you to write for you a meaningful letter of reference / recommendation. Better will be to develop relationships during subsequent block courses that can lead to meaningful letters.

Example solution scripts

Example solution scripts can be found here at the link below. However, please control your use of these example scripts carefully. Students in previous years have said they felt that they sometimes looked at the solutions to early or too quickly. They are provided because many students over many years have asked for them.

https://github.com/opetchey/BIO144/tree/master/6_example_scripts

Lecture podcasts

All lectures will be recorded and made available for later viewing.

Year specific information

2026 Detailed schedule

Lecture: Monday, 8:00-9:45am) Practical: Thurs or Fri. 1-3pm)

Week beginning	Content
Feb. 16	U1 Introduction, review, demonstration
Feb. 23	U2 R & RStudio
Mar. 2	U3 Regression part 1
Mar. 9	U4 Regression part 2
Mar. 16	U5 Analysis of variance
Mar. 23	U6 Multiple regression
Mar. 30	U7 Interactions
Apr. 6	– No lecture, no practicals
Apr. 13	U8 Count data
Apr. 20	– No lecture, review / catch-up practicals
Apr. 27	U9 Binary data
May 4	U10 Ordination
May 11	U11 Mixed models & What next?
May 18	U12 Review
May 25	– No lecture, review / catch-up practicals

2026 Locations

Lectures are in Y24-G-45
(Lectures will be live-streamed and podcasts will be made available, in any case.)

Practicals are in Y24-G-45.

2026 Assessment

Practice examination experience: Date and time to be communicated

Final Examination date and time: Please see this Studium Biology, Modulprufungen im Grundstudium web page and the links therein for the examination dates and other important examination information.

2026 Practice exam

We will share a practice exam with you to experience the exam platform and types of questions in the final exam. We likely make it available and notify you after the spring break.

2026 Exam viewing

If you did not get a 4.0 or higher:

You can view your final examination paper with Dr Frank Pennekamp - date, time and location to be announced by email.

You can view your resit examination paper with Dr Frank Pennekamp - date, time and location to be announced by email.

Learning Objectives

This document describes the learning objectives for **BIO144: Data Analysis in Biology**.

Course-level learning objectives

After successfully completing BIO144, students will be able to:

1. Formulate biological questions as statistical models, identifying appropriate response and explanatory variables.
2. Use R and RStudio to import, explore, visualise, and analyse biological data reproducibly.
3. Fit, interpret, and compare statistical models commonly used in biology, including linear models and generalised linear models.
4. Evaluate model assumptions, diagnose violations, and understand the consequences for inference.
5. Interpret model output quantitatively and biologically, rather than mechanically reporting p-values.
6. Communicate statistical results clearly, using figures, tables, and written explanations appropriate for biological audiences.
7. Read and critically assess quantitative analyses in published biological research.

Chapter-level learning objectives

Introduction

After this chapter, students will be able to:

- Explain the goals, structure, and expectations of the BIO144 course.
- Know the prior knowledge and skills required for success in the course.

- Understand appropriate use of AI tools in data analysis work and during the course.
- Describe a typical data analysis workflow, from developing question to communicating the answer.

R and RStudio

After this chapter, students will be able to:

- Navigate the RStudio interface and explain the purpose of scripts, the console, and the environment.
- Run R code and understand basic R syntax.
- Import data into R and inspect its structure.
- Know the many ways one can get help with R and RStudio.
- Understand what are and how to work with add-on packages in R.
- Perform basic data manipulation operations, including importing, viewing, and various manipulations.
- Produce simple exploratory data visualisations.
- Use scripts to support reproducible data analysis workflows.

Simple linear regression – Part 1

After this chapter, students will be able to:

- Explain the purpose of simple linear regression in biological data analysis.
- Understand the mathematical form of a simple linear regression model.
- Describe how the slope and intercept are calculated.
- Describe how the error / variation is modelled.
- State the assumptions underlying simple linear regression.
- Assess if the assumptions are reasonably met using diagnostic plots.
- Recognise and remedy common problems encountered when fitting linear regression models.
- Perform linear regression in R.

Simple linear regression – Part 2

After this chapter, students will be able to:

- Understand how to measure how good is the regression (correlation and R-squared).
- Test and explain if the parameter estimates are compatible with some specific value (t-test).

- Understand how to find the range of parameters values are compatible with the data (confidence intervals).
- Understand what are the regression lines compatible with the data (confidence band), and how to construct these in R.
- Understand what are the plausible values of newly collected data (prediction band) and how to calculate these in R.
- Interpret the biological meaning of regression coefficients.
- Know how to communicate the results of a linear regression analysis effectively.
- Critically assess the strength and limitations of linear regression models.

One-way ANOVA

After this chapter, students will be able to:

- Explain how ANOVA fits within the linear model framework.
- Fit a one-way ANOVA model using `lm()`.
- Interpret group means and differences between groups.
- Explain the concept of variance partitioning.
- Understand the connection between ANOVA and regression with categorical predictors.
- Decide when ANOVA is an appropriate modelling approach for biological data.
- Perform linear regression in R, assess model assumptions, and interpret results in a biological context.
- Communicate ANOVA results effectively using text, tables, and visualisations.

Multiple regression

After this chapter, students will be able to:

- Understand what is a question that multiple regression can help answer.
- Know the mathematical form of multiple regression models.
- Fit linear models with multiple explanatory variables.
- Assess if model assumptions are reasonably met using diagnostic plots.
- Assess if the group of explanatory variables significantly explain variation in the response.
- Describe which variables are associated with the response, and the direction of these associations.
- Measure the amount of variation explained by the model (R-squared, adjusted R-squared).
- Assess the relative importance of different explanatory variables.
- Make conditional predictions from multiple regression models.

- Understand what is collinearity, and how it affects multiple regression.
- Use R to fit, diagnose, and interpret multiple regression models.
- Communicate multiple regression results effectively using text, tables, and visualisations.

Interactions

After this chapter, students will be able to:

- Explain what an interaction between explanatory variables means biologically and mathematically.
- Fit models that include interaction terms.
- Interpret interaction coefficients correctly.
- Visualise and explain how effects depend on the values of other variables.
- Distinguish additive from non-additive biological effects.
- Judge when interactions are biologically meaningful and justified.
- Understand interactions in the context of both categorical and continuous explanatory variables, ANCOVA, two-way ANOVA, and multiple regression.
- Communicate results involving interactions clearly and accurately.
- Use R to fit, diagnose, interpret, and communicate models with interactions.

Generalised linear models for count data

After this chapter, students will be able to:

- Describe the kinds of biological questions and processes that involve count data.
- Recognise when, why, and how count data violate the assumptions of linear models.
- Explain why normal error assumptions are inappropriate for count data.
- Describe the key components of a generalised linear model.
- Fit Poisson regression models using `glm()`.
- Interpret model coefficients on the appropriate scale.
- Assess whether a Poisson model provides an adequate description of the data.
- Use R to fit, diagnose, and interpret GLMs for count data.
- Communicate results from count data analyses effectively.

Generalised linear models for binary data

After this chapter, students will be able to:

- Identify binary response variables in biological datasets, and be familiar with common examples (e.g., presence–absence, success–failure) and the types of questions they can arise from.
- Explain why linear models are unsuitable for binary outcomes.
- Fit logistic regression models using `glm()`.
- Interpret model coefficients in terms of probabilities and odds.
- Visualise fitted relationships for binary data.
- Understand how logistic regression supports biological inference about presence–absence or success–failure outcomes.
- Use R to fit, diagnose, and interpret GLMs for binary data.
- Communicate results from binary data analyses effectively.

Ordination and multivariate data

After this chapter, students will be able to:

- Recognise when biological questions involve many response variables simultaneously.
- Explain the goals of ordination methods.
- Interpret ordination plots in terms of similarities and differences among observations.
- Understand ordination as a tool for dimensionality reduction.
- Critically assess what information ordination methods do and do not provide.
- Use R to perform basic ordination analyses (i.e., PCA, NMDS).
- Communicate ordination results effectively using visualisations and text.

Mixed models and what next

Part 1: Mixed models

After completing this part, students will be able to:

- Recognise biological situations in which data are **grouped, nested, or hierarchical**.
- Explain why standard linear models may be inappropriate for grouped data.
- Describe the conceptual difference between **fixed effects** and **random effects**.
- Understand random effects as a way of modelling **structured sources of variation**.
- Interpret mixed models as extensions of linear models, rather than as fundamentally different tools.

- Identify common biological examples where mixed models are appropriate (e.g. repeated measures, individuals within populations, sites within regions).
- Interpret mixed model output at a conceptual level, focusing on biological meaning rather than technical detail.
- Appreciate the role of **partial pooling** in balancing information across groups.
- Recognise the limitations and assumptions of mixed models without needing to master implementation details.
- Use R to fit simple linear mixed models using the `lme4` package.
- Communicate results from mixed model analyses effectively, focusing on biological interpretation.

Part 2: What next?

After completing this part, students will be able to:

- Place the statistical models learned in BIO144 within a broader landscape of data analysis methods.
- Recognise when more advanced models may be required to address biological questions.
- Identify directions for further learning in statistics and data analysis.
- Understand that statistical modelling is an **iterative and evolving process**, not a fixed set of rules.
- Reflect critically on the limits of the models used in the course.
- Appreciate the importance of biological reasoning alongside statistical tools.
- Approach future quantitative methods courses and analyses with confidence and curiosity.

Review (L12)

No additional learning objectives.

FAQs

Organization-related FAQ

How to ask a question so we can easily help?

You're probably quite competent at asking questions so that we can easily help. But here are a few tips that can help you be even better:

- If you're asking a question in the OLAT Forum and include R code, please try to ensure it is easy to read and copy into R.
- If you're asking for help about some code that doesn't work, please consider making a "minimal, reproducible example". This web page describes what one is and how to make one (https://stackoverflow.com/help/minimal-reproducible-example).

Is practical attendance mandatory?

You are free to do the practicals whenever you want, but we encourage that you do attend the practical afternoons, especially if you have questions!

Can I change which practical afternoon I attend?

In general it's better that you stick to the afternoon you registered for, but if you cannot attend it you can of course join the practical on the other day.

Why is the number of attempts for the questions limited?

For now, the tests are implemented such that you can take each test **only** once, but you can suspend every test you take. This saves your progress within the test and when you get back to it you'll pick up exactly where you left it (your progress is saved). You can only take the tests once, because if you can take it more than once your progress is lost, meaning that you would have to retake even the questions that you already answered correctly (OLAT does not allow us to change this).

However, within each test attempt, you get **2 tries per question**:

- For the weekly graded assessments there must be a limited number of tries for the graded assessment questions.
- For self-tests in the homework and practical parts this is not set in stone. For now, for almost all questions you get 2 tries because giving more attempts might demotivate some people. This might change in the future.

Are the sections named Extras part of the exam?

No, they are just for your interest!

Device with enough battery for exam

If you are not certain that the battery of your laptop can last for the duration of the final examination, please let us know using the form at the end of the course content here on OLAT.

About the Discussion Forum

Use the discussion forum to ask questions, give tips, make suggestions about how the course could be improved… or anything else that springs to mind while you're taking the course.

However, before posting please check out the Index in the "Wiki: FAQ" section in this course to see whether your question has already been answered there and also look at the other forum posts here to see whether there is already a discussion thread for your topic. If you cannot find what you are looking for, go ahead and post something in this forum, we will respond as soon as possible!

Other potential uses of the forum are that you may make a list of questions that you want to ask during the class, or a list of things that you'd like us to go over again in class, or in more detail in lectures or practicals. Then we can look at these in lectures or practicals.

One rule: do not post answers to assessment or solutions to exercises. Help others find and realise for themselves the answer, but don't post the actual answer. If you post comments/questions about graded assessments, we will always investigate and take appropriate action. This may include deleting or editing your post. Please ask about GAs during a practical or email.

Another rule / guideline. Open a new post / thread for a new issue.

R-related FAQ

What version of R is needed?

It is good practice to keep R up to date, so it is recommended that you have the most recent version of R installed. In any case, you should at least have the R version 4.0 or newer installed. You can check the version by running "version" or "sessionInfo()" in R.

What packages are needed?

Note: this list is incomplete and will be updated in the future.

The packages you need in each unit are usually in the lecture and/or in the homework material.

An incomplete list (work in progress):

- tidyverse (contains ggplot2, readr, dplyr, tidyr, etc.)
- skimr
- ggfortify
- …

I am having issues installing a package

Problems with installing packages are somewhat common. Often, these problems be avoided by using

```
install.packages( "NAME", type = "binary")
```

instead of simply

```
install.packages( "NAME", )
```

The reason is that quite often, packages are available a source packages, which must be compiled locally, which is often not possible on standard R machines. Therefore, by specifying type = "binary", you tell R to install the already compiled version, even though it might be an older version.

Different output of summary()/str()/etc. function than what is shown in video

During the course it will happen that the output that you get from function such as summary(), glimpse(), str(), …, will look a bit different from the output that you see in the (homework) videos.

Usually, the reason for this is that since R version 4.0.0 (and newer) the function read.csv() no longer reads in non-numerical variables as factor variables, but instead these variables are read in as character variables and functions such as summary(), glimpse(), str(), … handle character variables differently than they handle factor variables.

What does rm(list=ls()) do?

While working in R we assign stuff to variables, load in data, etc. These things are stored in the R "environment". The line rm(list=ls()) deletes everything that is stored in this environment. The reason for this is basically that by freeing up space we are on top of things (when many different things are stored in the environment it can get messy) and are sure that we are not using things from (for instance) a previously run script.

The function rm() stand for “remove” so it removes everything that we give it. In this case we give it a list of things: ls() lists every name that is stored in the environment. We usually use it as the first thing in a script.

How do I set the number of decimal places shown?

Use the round() function to tell R how many decimal places to report and what rounding to do. You otherwise cannot know for sure how a number displayed in the console has been rounded or not.

What does the predict() function do?

In general, the predict() function takes as argument a model object, for instance one fitted with lm(). It then predicts/estimates/calculates the response variable based on the values of the covariates (explanatory variables) and the estimated coefficients according to the fitted model. You can do this by hand (good for understanding).

If you do not specify the data to use for the prediction, it will automatically use the data that was used in fitting the model. You can also specify the dataset used, which is useful, for instance when you have multiple covariates and you want to visualize the relation of one specific explanatory variable with the response variable. In that case you create a new dataset in which you keep every covariate at constant values (e.g. their mean) with the exception for the one in you are interested in. Then you can use this dataset in the predict() function and then, if you want, you can plot the relation of that variable with the response.

Note: predict() does **not** fit a model, it evaluates a model with some given data

What does the expand.grid() function do?

The expand.grid() function is a handy function to create new data, as it returns every possible combinations of the input provided. For instance expand.grid(a = 1:2, b = 4:5) returns

```
    a b
1   1 4
2   2 4
3   1 5
4   2 5
```

We often use this function to create new data because we want to visualize the effect of one particular variable on the response variable according to the fitted model. When we have a model with more than 1 explanatory variable, these variables can vary together and if they do the effect of one particular variable on the response variable is more difficult to see. Hence to best show the importance of a particular variable, we keep all other variables at constant values. These constant values should ideally be sensible values, i.e. keeping the mass at 0 does not make sense, but it makes more sense to keep at the mean value.

AIC(model) versus extractAIC(model)

You can find an explanation of the difference here (see the accepted answer)

It does not matter which of the two functions you use, but be sure to always use the same function within an analysis/script!

The autoplot function does not show the leverage plot

When there is one categorical and one continuous variable (and in other cases as well), the forth plot shown by the autoplot() function (in the ggfortify package) is not the leverage plot that we would usually get. See also here. It has to do with how this function is implemented.

If you want to see the usual leverage plot, see the above link to see how to achieve that.

The other three plots shown by the function are more important, so it is ok that we do not see the leverage plot sometimes.

Cooks distance in leverage plot

I am not aware that autoplot() has a method to show Cook's distance. This is not so important. The important thing about that plot is to see whether there are data points that have both a large leverage (a big impact on the regression fit result) and large residual value. These points would be in the top and bottom right corner of the fourth plot. If there are such points it can be interesting to repeat the analysis without these points to see how the result changes. Note that this plot is not needed to check the model assumptions, so for now at least the other three plots are more important.

gather() vs pivot_longer() and spread() vs pivot_wider()

This course was written when `gather` and `spread` were still the go-to `tidyr` functions for these tasks (there are base `R` functions for them as well). Since then, `pivot\_wider` and `pivot\_longer` have been released, which have more functionalities and are more flexible. For such simple tasks both work perfectly fine, with the advantage that `spread` and `gather` have slightly more intuitive names and are those more easily understood. But of course, as recommended in the help page, the newer functions should be used moving forward (at some point this course will switch as well).

unique() vs levels()

The function `levels()` only works with factors. In newer versions of `R` variables are read in by `read\_csv` as characters instead of factors (see also here). Use the function `unique()` instead of `levels()`

Statistics-related FAQ

What is the difference between fitted and predicted values?

Fitted values: The response values you get when you use the observed values of the explanatory variables, i.e. the ones you used to fit the model: the original data

Predicted values: In general, the response values you get when you use new observations of the explanatory variables (i.e. new data). Note: if the new data is the same as the original data, then predicted values == fitted values.

This being said, it happens that the terms fitted values and predicted values are used interchangeably, even though there is a (small) difference.

Interpretation of p-values

The p-value is best interpreted as:

The p-value is the probability of having a test statistics value at least as extreme as the value actually observed (e.g. the t-value of the t-test or F-value of the F-test) given that the Null hypothesis is correct (often H0: parameter=0). If the p-value is below 0.05 we reject H0 and go with H1 (often H1: parameter =/= 0). The p-value is **not** the probability of H0 being correct.

See also the lecture material.

Interpretation of confidence intervals

Let's assume we have a simple linear regression with body-fat as the response and BMI as the explanatory variable. We find that the slope of BMI is 1.75 with a corresponding 95% confidence interval [1.55,1.97]. This is interpreted as follows:

- "All true values for beta1 (slope for BMI) between 1.55 and 1.97 are compatible with the observed data".
- To rephrase this slightly, this means: the **true** slope for BMI could have any value between 1.55 and 1.97 and we would not be surprised to see the data that we have (and thus, if the **true** slope was outside the interval, it would be surprising to see the data that we have: the estimated slope based on the data does not match the **true** slope.

Note: I want to stress that it is **not correct** to say that there is a 95% probability that the **true** slope is between 1.55 and 1.96 (see below).

So, with the interpretation given above, if the confidence interval of the slope does not cover 0, then a slope of 0 is not compatible with the data. In fact, in the case of the 95% confidence interval this means that the p-value is below 0.05, which means that we reject the Null hypothesis H0 that the slope for BMI is 0. **Note**: this is neither "good" nor "bad", it is just what it is: we are just interested in the truth.

There is an alternative interpretation of confidence intervals which is arguably more precise (but you can use the above for this course). To explain it, here are a couple of things that I personally think are good to keep in mind:

- The parameter of interest (e.g. the slope of BMI) is estimated based on the available data, meaning that the estimate is a function of the data. If for instance the experiment is repeated the data will look a little bit different and the estimate will be a little bit different. However: the true value of the parameter of interest (the slope) is the same in both experiments!
- The same thing goes for confidence intervals: they are estimated based on the data, so if the data changes a bit, the upper and the lower bound of the confidence interval will be a bit different as well.
- An example of wrong interpretation: continuing the example above, let's say we know the true slope of BMI and it is 2. What is the probability that the confidence interval includes the **true** slope? 0 (!), because 2 is not in [1.55,1.97]. And what would the probability be if the **true** slope is 1.6? 1 (!), because 1.6 is included in [1.55,1.97]. **This is why it is incorrect to say that there is a 95% probability that the estimated confidence interval contains the true slope**: the **estimated** confidence interval either contains or it does not contain the true slope.

Now, because of probability theory reasons, a confidence interval is best interpreted like this:

- "If we repeat the entire experiment many times (and thus collect slightly different data each time) and for each experiment we calculate the 95% confidence interval, then 95% of the calculated confidence intervals will contain the true parameter value".
- This definition is a bit harder to get, but it basically tells us that it is a measure of uncertainty of the estimated slope.

Model diagnostics residuals versus fitted values

This entry is not complete

The central assumption of linear regression is that the errors are independent (from each other) and identically distributed (i.i.d.) following a normal distribution with mean = 0 and variance = sigma^2. We cannot measure the errors directly, but we have estimates of them: the residuals. This means that we can check these assumptions based on the residuals, because given that they are estimates of the errors and that the model is valid the residuals have to be i.i.d. following a normal distribution as well. In the figure residuals vs. fitted values we can check for some of these assumptions:

- Mean = 0: the mean has to be zero across all fitted values. So the points in the figure have to be spread approximately symmetrically above and below 0, so that the mean is approximately 0 across all fitted values. If for instance we see that for larger fitted values the mean of the residuals increases or decreases clearly away from 0, then this assumption might be violated.
- Variance == sigma^2: constant variance. Similarly to above, the variance has to be independent from the fitted values, meaning that the spread of point below and above 0 needs to be approximately the same across all fitted values. If this is not the case this results in a pattern (e.g. larger fitted values result in a higher variance in the residuals) and this assumption might be violated.

Difference between binomial and count data

For count data we use Poisson regression and for binomial data we use binomial regression (commonly called logistic regression)

- Count data come from questions like "how many times have you voted in your life?" and the answer can be a positive integer (including 0), while binary data come from questions like "did you vote last year?" to which the answer can only be yes (1) or no (0).
- We get binomial data if we have n (=number of persons) independent Bernoulli trials (i.e. binary trials), so we would basically count how many people have voted last year out of the n we asked. In other words, if we have 7 yes and 3 no to the question "did you vote last year?", they come from 10 different people (sample size n=10), while if to the question "how many times have you voted in your life?" we get an answer like "I've voted 10 times in my life" this comes from a single person (sample size n=1), so keep this in mind.

Dispersion parameter

Why is the value of the estimated dispersion parameter different to residual deviation divided by the residual degrees of freedom?

The actual dispersion is based on a weighted version of the residuals:

object is the output of a glm(… "pseudopoisson") call:

dispersion <- sum(object$weights * object$residuals^2)/object$df.residual

residuals are not: